Big Footprints Point to a Giant Mammal in Dinosaur Times

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Picture a moonlit night, about 130 million years ago, in what is now southern Brazil. A large mammal walks across wet sand **dunes**, leaving tracks five inches wide.

Quarry workers found those **fossil** footprints in 2019. Scientists still do not know which animal made them. But the prints are nearly twice as big as the next-biggest mammal tracks from the same place. The animal walked on all fours, with short legs and paws, and was about the size of a **cougar**.

The find goes against an old idea. Since the 1800s, many scientists thought mammals stayed small while dinosaurs ruled. But other clues show some early mammals grew large. In Poland, bones of **Lisowicia**, a pig-like plant-eater as big as an elephant from about 205 million years ago, show an animal over eight feet tall and heavier than five tons.

Most mammals in dinosaur times were still small, often smaller than a house cat. But the **fossil** record is not complete, and more big mammals may be waiting to be found.

Word Treasure

fossil -- A fossil is the saved mark or remains of a living thing from long ago.
dunes -- A dune is a hill of sand made by wind.
Quarry -- A quarry is a place where workers dig stone out of the ground.
cougar -- A cougar is a large wild cat.
Lisowicia -- Lisowicia was a pig-like plant-eater from long ago.

Background you need

Long ago, dinosaurs lived alongside early mammals, which were mostly small animals with fur that fed their babies milk.

When an animal walked on soft ground, its tracks could harden into rock and last as fossils, so scientists can study them to learn how big the animal was and how it moved.

Finding very large tracks can change what scientists think about how big early mammals could grow.

Break it down

WHO: Quarry workers who found the footprints, and scientists studying them
WHAT: Fossil footprints five inches wide left by a large mammal about the size of a cougar
WHERE: In what is now southern Brazil, on wet sand dunes
WHY: The big tracks show some early mammals grew large, going against the old idea that mammals stayed small while dinosaurs ruled

Why it matters

It shows that what scientists believe can change when new clues turn up, just like when you find a new clue that changes your answer to a puzzle.

Test what you remember

1. Where were the fossil footprints found?

- (A) On the Moon
- (B) In southern Brazil
- (C) In Poland
- (D) In a zoo

2. How wide were the tracks?

- (A) As wide as a house
- (B) One inch wide
- (C) Five inches wide
- (D) Ten feet wide

3. What size was the animal, compared with a modern animal?

- (A) About the size of a house cat
- (B) About the size of a mouse
- (C) About the size of an elephant
- (D) About the size of a cougar

4. What did scientists once think about mammals in dinosaur times?

- (A) They lived only in Poland
- (B) They stayed small while dinosaurs ruled
- (C) They were bigger than dinosaurs
- (D) They could fly

5. What was Lisowicia?

- (A) A small flying mammal
- (B) A quarry worker
- (C) A kind of dinosaur
- (D) A pig-like plant-eater as big as an elephant

6. What do scientists still NOT know about the footprints?

- (A) Where they were found
- (B) How wide they are
- (C) Which animal made them
- (D) When they were found

Answer key (try the quiz first!): 1: B 2: C 3: D 4: B 5: D 6: C

Questions to think about

1. Many scientists in the past: They thought mammals stayed small while dinosaurs ruled.

MY THOUGHTS (at least 20 words):